

AI-compute supply chain: chart pack

Current state of the analysis. Data: UN Comtrade + TDM monthly panel (60 HS6 codes, 2017-01..2026-04, checked against Atlas) and Atlas annual data. Sections: 1 the chain as flows (2024) -- 2 the chain in stages -- 3 the factor model over time -- 4 concentration & fragmentation (sections 3 and 4 use the AI-compute codes only). Details: docs/data.md, docs/supply-chain-narrative.md, results/.

2026-08-20

1. The chain as flows

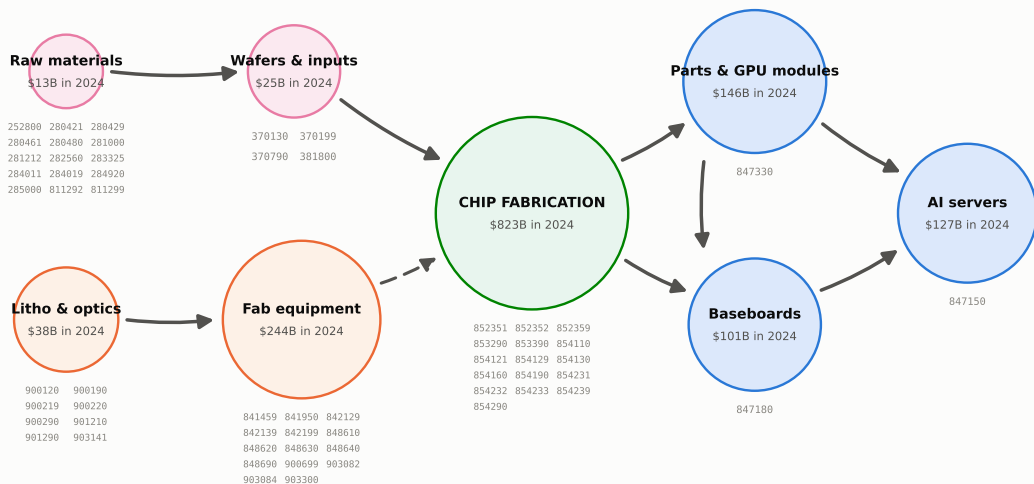
2024 trade flows. China and Hong Kong are counted as one (CHK) throughout. Stages 1-5 use annual Atlas data; stages 6-8 our monthly panel.

1. The chain as flows

The shape of the chain

Two input branches feed chip making; finished hardware flows out to the right. Pink inputs are used up per chip; orange equipment builds capacity. Blue: the three codes of our monthly panel. Circle size grows with the log of 2024 trade.

The AI-compute supply chain — stylized shape, with the HS6 codes in each node



Two input branches, one output line. Materials (pink) -- silicon, chemicals, wafers -- are used up with every chip made (solid arrows). Equipment (orange) -- lithography and other fab tools -- is investment in capacity: in Taiwan and Korea, equipment imports rise six to twelve months before chip exports (dashed arrow). Korean memory and Taiwanese chips are packaged together, usually in Taiwan. Chip design earns as a service and crosses no border. Circle diameters grow with the log of 2024 world trade. Blue marks the three codes of our monthly panel (the Fed AI-compute basket). A few niche inputs (lasers, vacuum pumps and valves, chip substrates) fall outside the 60 codes.

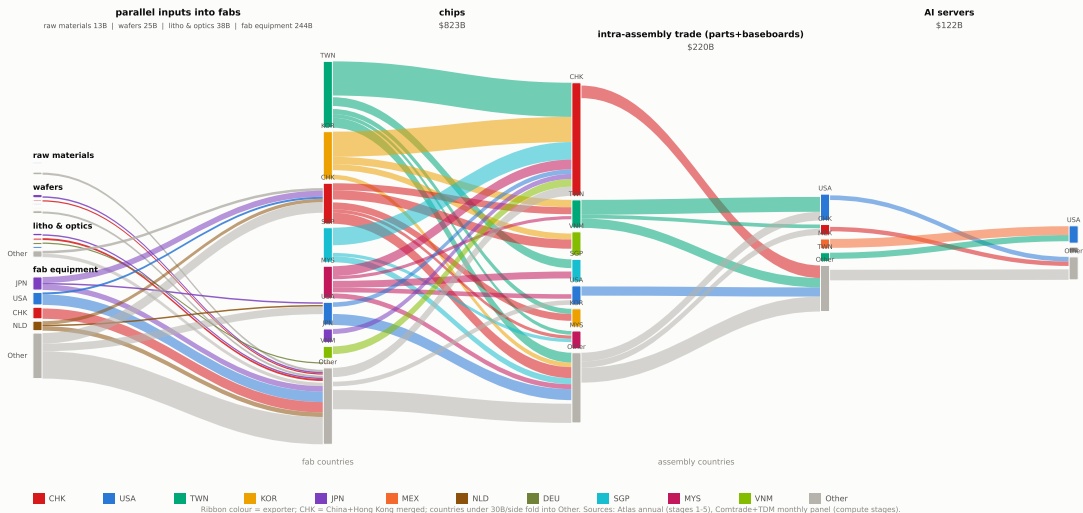
Codes: OECD semiconductor mapping + Fed basket (docs/data.md, section 1).

1. The chain as flows

The whole chain on one dollar scale

Band width = 2024 dollars. The chips stage dominates; small input flows are drawn but too thin to see.

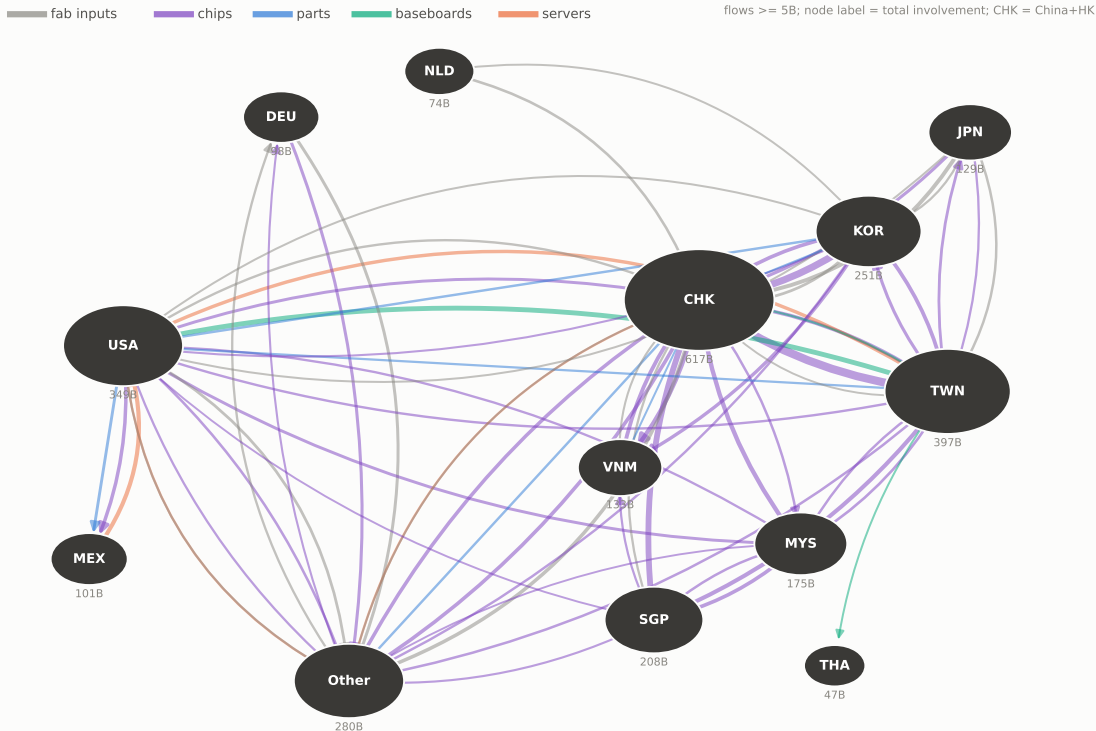
The AI-compute supply chain, 2024 — parallel inputs converge on the fabs; chips flow on to servers (one dollar scale — band widths nominally comparable everywhere)



The same flows as a country map

Countries as nodes, flows as edges colored by stage. Shows combined roles: Mexico takes in chips and parts and ships out servers.

The AI-compute supply chain as a country network, 2024 — edges by product stage, width ~ \$



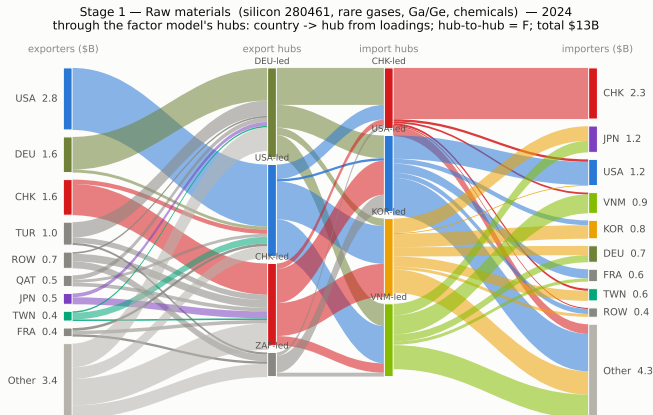
2. The chain in stages

One page per stage: exporters on the left, importers on the right, with the model's groupings in between.
Groupings fit the 2024 flows closely (R-squared 0.95-0.99).

2. The chain in stages

Stage 1 -- raw materials

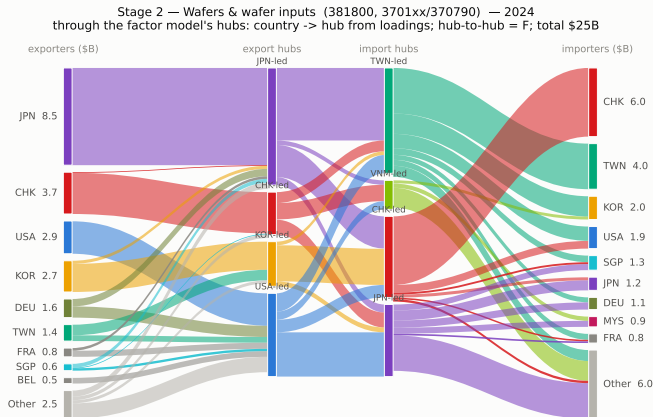
Reading guide for all stage pages: left = exporters, right = importers, middle = the model's export and import groups. Much raw-materials trade is not chip-related (the codes also cover welding gas, abrasives, fertilizer inputs), so flows scatter widely.



2. The chain in stages

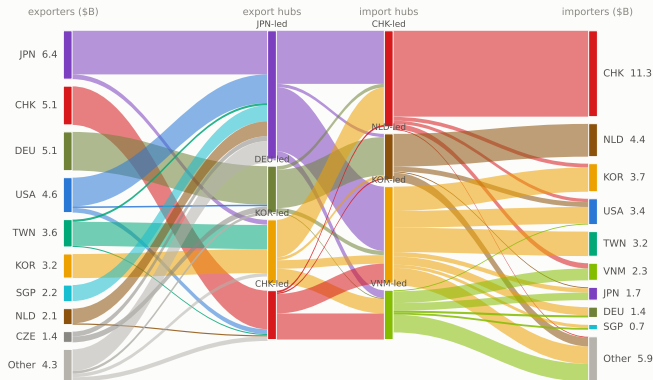
Stage 2 -- wafers and wafer inputs

Japan dominates the export side.



Stage 3 -- lithography and optics inputs

Stage 3 — Lithography & optics inputs (9001xx/9002xx, 901210/90, 903141) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$38B

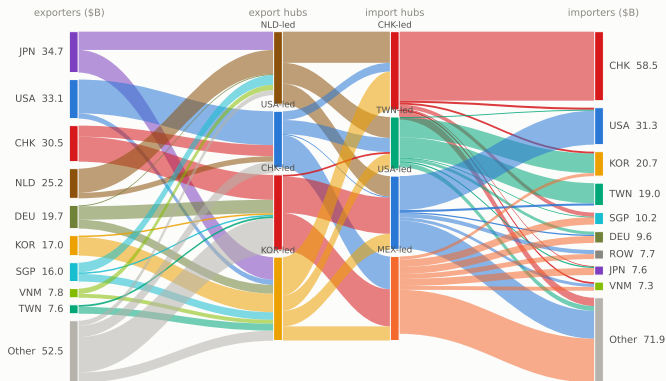


2. The chain in stages

Stage 4 -- fab equipment

Machines, not materials: countries buy these to build capacity, and their imports rise 6-12 months before chip exports.

Stage 4 — Fab equipment (8486xx, metrology, fab plant) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$244B



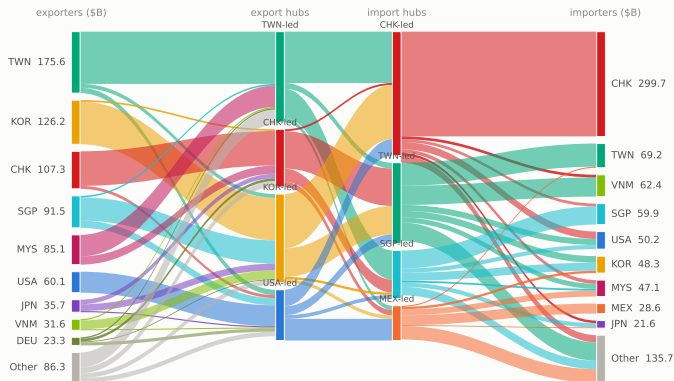
Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 (no monthly panel upstream)); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral.

2. The chain in stages

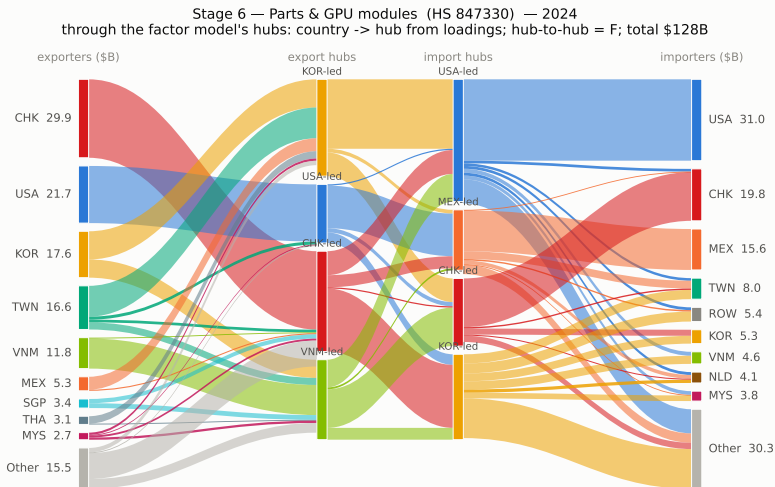
Stage 5 -- chips

The biggest stage, and the whole electronics industry -- phones, cars and PCs, not just AI. Most chips flow to Asian assembly; the China sphere absorbs about 43%.

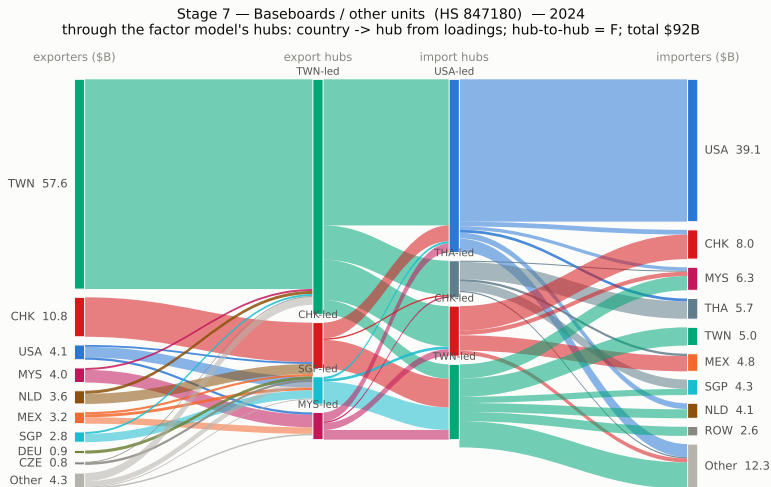
Stage 5 — Chips (logic, memory, discretes) (8542xx, 8541xx, media) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$823B



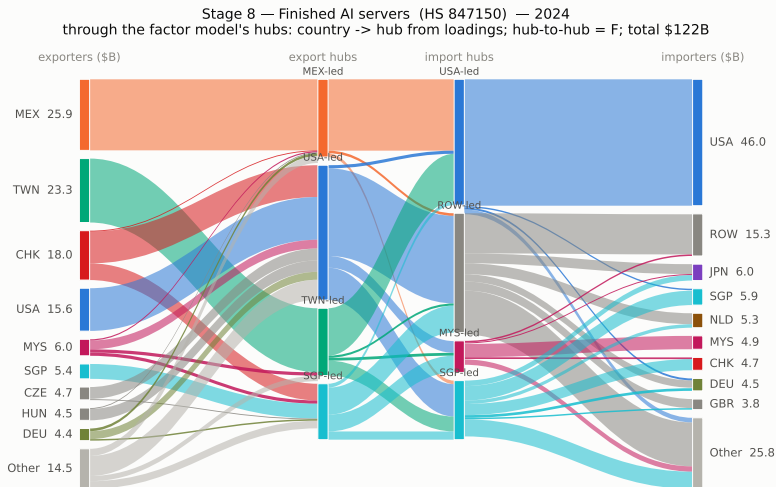
Stage 6 -- parts and GPU modules (847330)



Stage 7 -- baseboards (847180)



Stage 8 -- AI servers (847150)



factor model over time -- AI-compute code

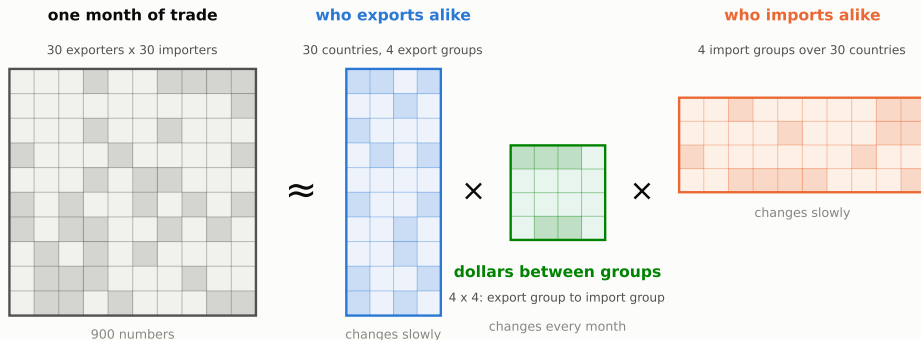
This section uses only the three blue codes (parts/GPU modules, baseboards, AI servers), monthly from 2017. Unless the page says otherwise, China and Hong Kong are counted as one. Shaded bands mark the periods between structural breaks.

3. The factor model over time -- AI-compute codes only

How the model works

The one idea behind every page that follows.

How the factor model summarizes the trade data

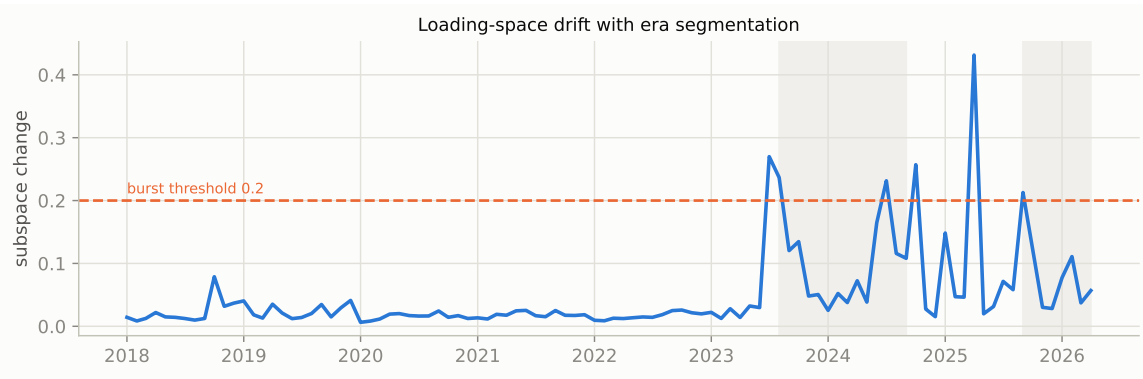


Every month, 900 country-to-country flows are compressed into: groups of exporters that move together, groups of importers, and a small table of dollars between the groups. The groups are the slow structure of the network -- when they shift, the model marks a break. The small table is the fast part: its cells are the trade channels, and three of them carry the AI boom.

3. The factor model over time -- AI-compute codes only

When the network changed shape

The line measures how much the export groupings moved from one month to the next. Flat for six years, then: Aug 2023 (AI demand), Oct 2024 (the quarter memory export controls were signaled), Sep 2025.

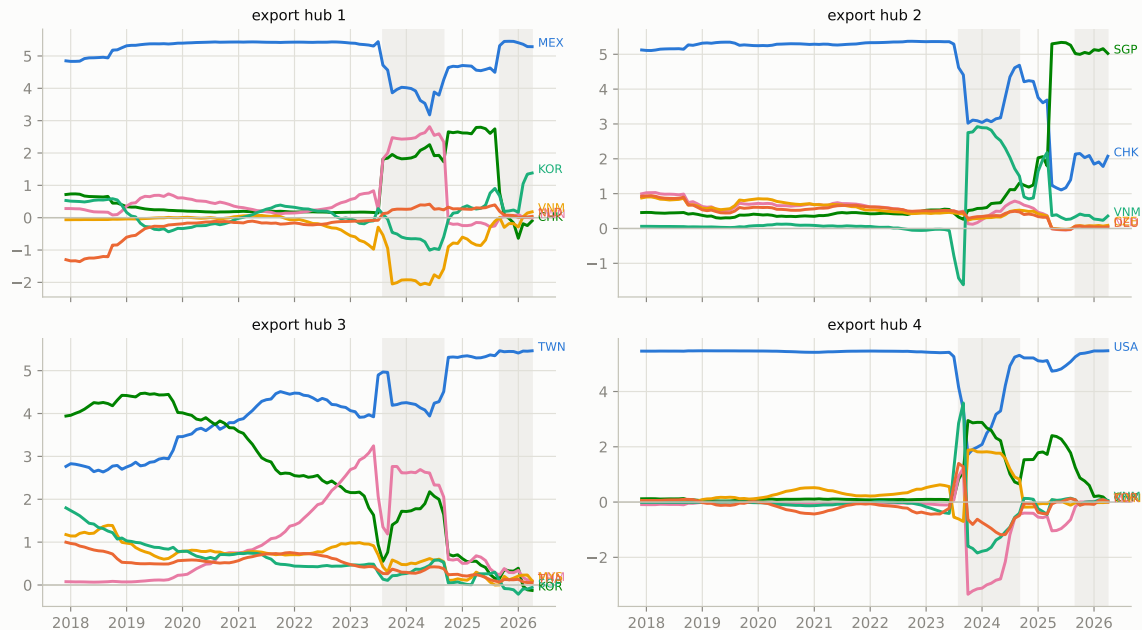


3. The factor model over time -- AI-compute codes only

Who is in each export group

At Oct 2024 the China group loses its separate identity and a Singapore-led group appears.

AI compute, China+HKG bloc (TWN separate) — export loadings, era-anchored (12m window; shaded bands = eras)

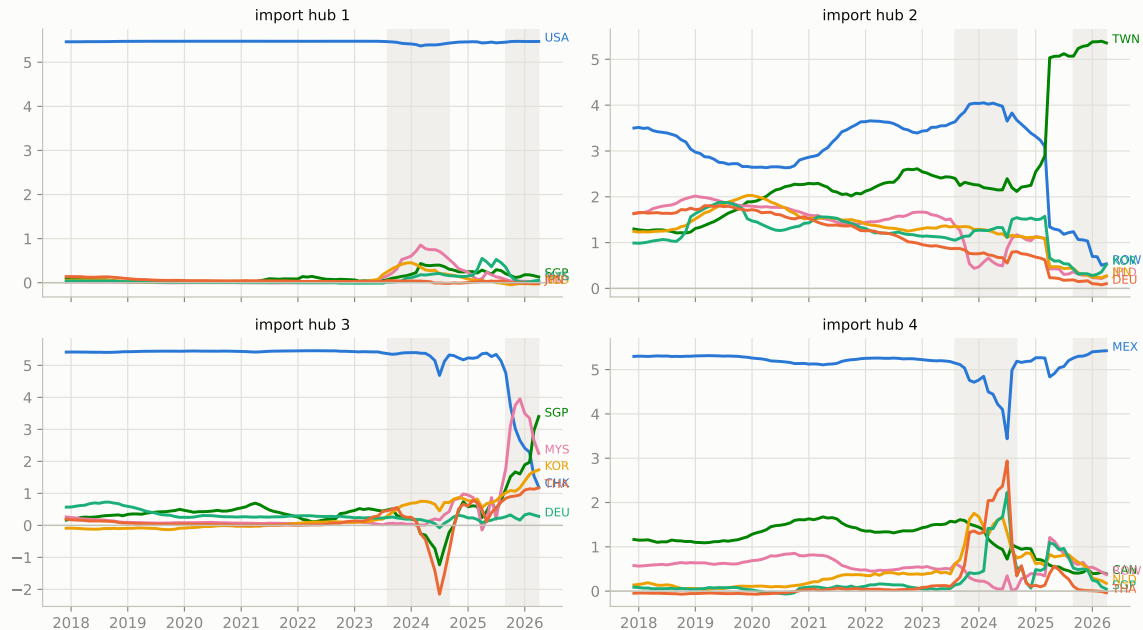


3. The factor model over time -- AI-compute codes only

Who is in each import group

The import side is steady: a US group, a Mexico group, a Taiwan group, and the rest of the world.

AI compute, China+HKG bloc (TWN separate) — import loadings, era-anchored (12m window; shaded bands = eras)

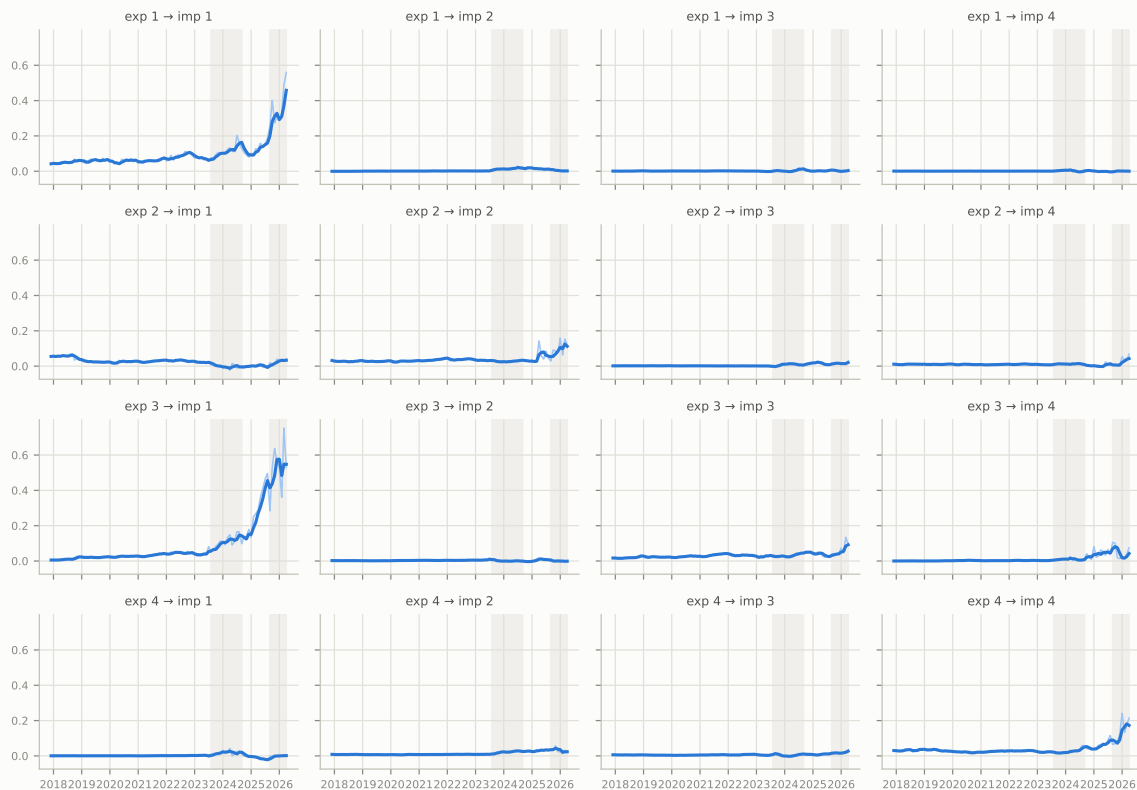


3. The factor model over time -- AI-compute codes only

Dollars between groups, month by month

Three of the sixteen channels carry the AI boom: Taiwan to the US, and both legs of the US-Mexico assembly loop. The rest are flat.

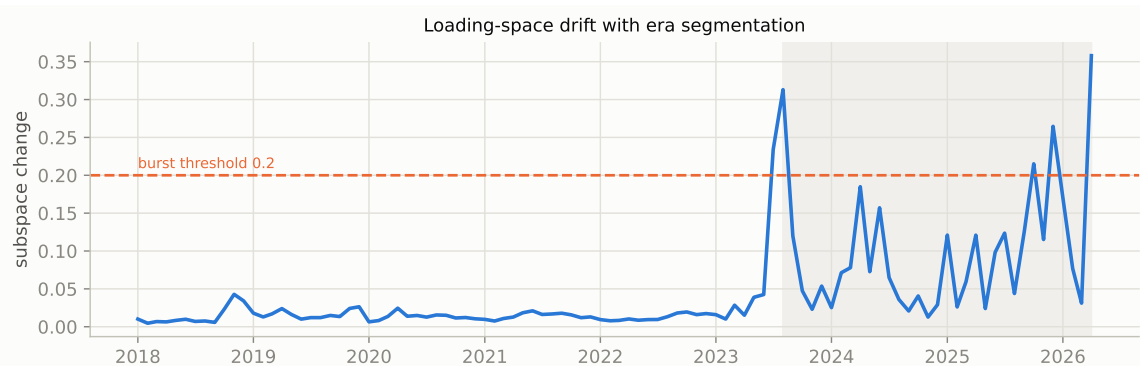
AI compute, China+HKG bloc (TWN separate) — hub-to-hub F_t , era-anchored (3m MA thick)



3. The factor model over time -- AI-compute codes only

Same, with every country separate

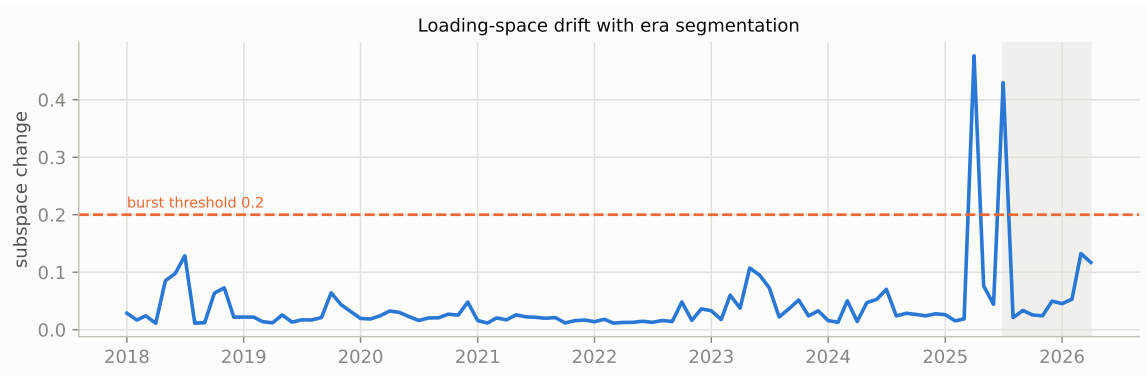
Without merging China and Hong Kong there is one break in nine years: Aug 2023. Demand changed the country-level structure; policy did not.



3. The factor model over time -- AI-compute codes only

Parts only (847330)

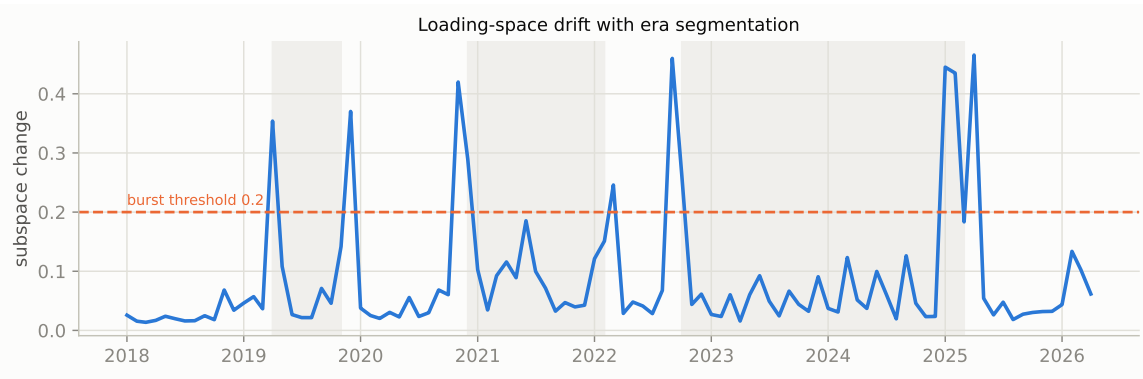
91 months without a break -- the stable base layer of the network.



3. The factor model over time -- AI-compute codes only

With both blocs merged

Merging China+Hong Kong and US+Mexico hides churn inside each bloc and isolates what happens between them. The breaks line up with policy: the 2019 tariff war, the Oct 2022 export controls, the Apr 2025 tariffs.

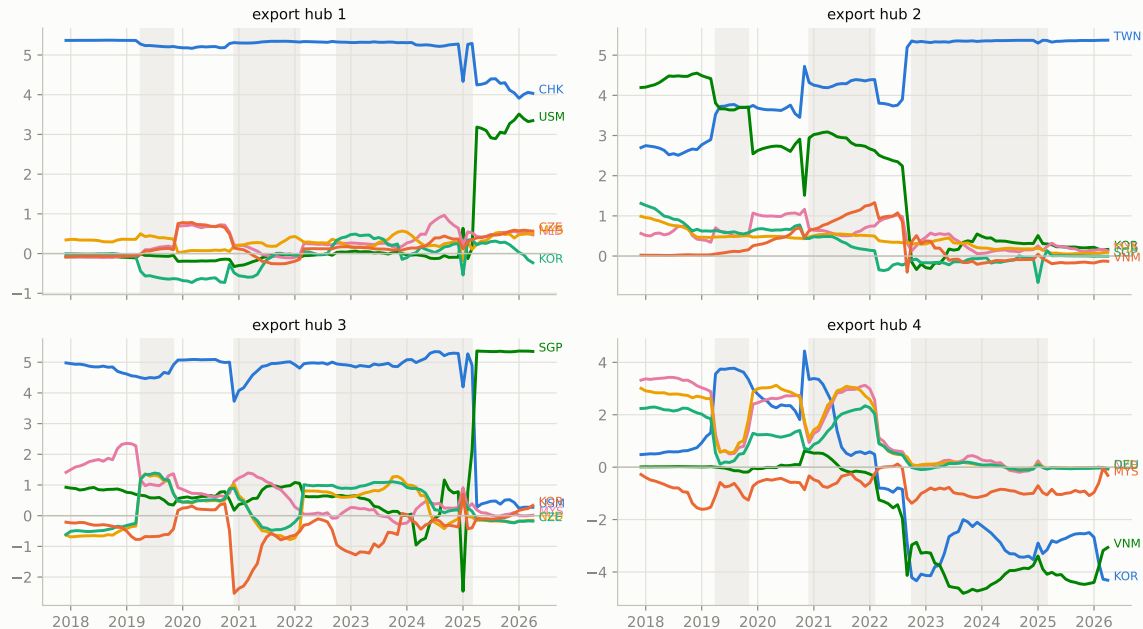


3. The factor model over time -- AI-compute codes only

The tariff signature

After Apr 2025 the two blocs move together on a single factor
-- seen at no other break.

AI compute, CHN+HKG and USA+MEX blocs — export loadings, era-anchored (12m window; shaded bands = eras)



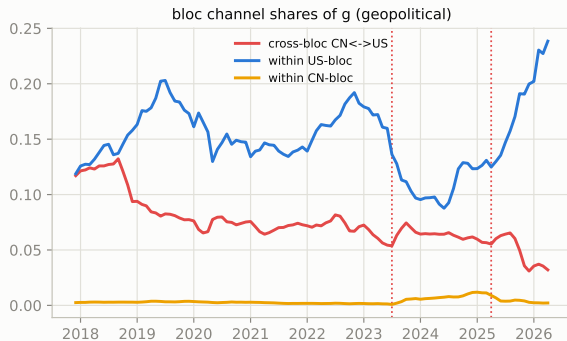
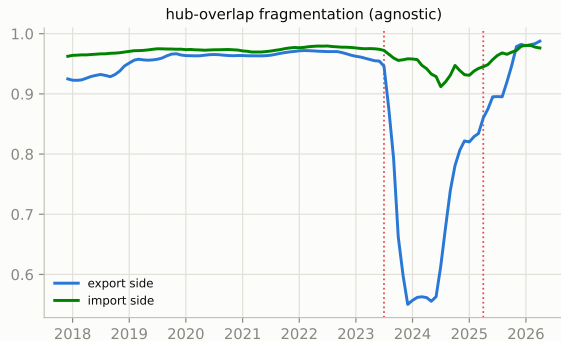
Concentration & fragmentation -- AI-compute co

Monthly measures built on top of the model, for the three blue codes only. Dotted lines mark Jul 2023 and Apr 2025.
Interpretation: [results/network_stats/findings.md](#).

Two meanings of fragmentation

Left: do the model's groups overlap (a technical check, little economic signal). Right: the share of trade crossing between the China and US blocs -- down about 70% since 2017, in two steps (2019, then 2023 on), with no recovery between.

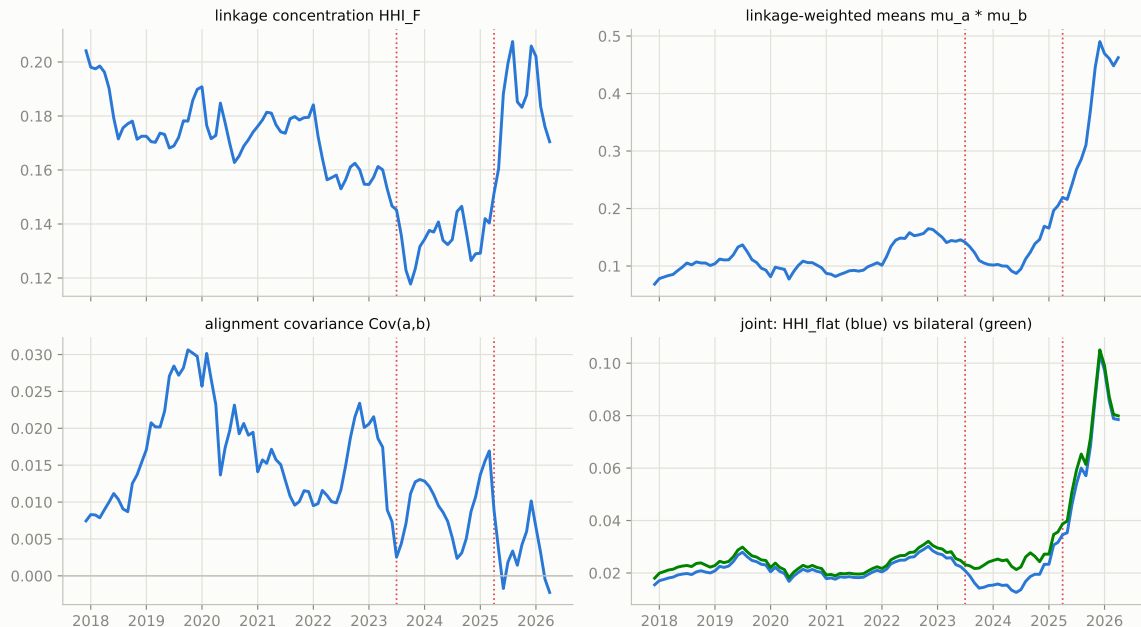
AI compute (3 codes) — two notions of fragmentation (3m MA)



Concentration, taken apart

Overall concentration splits exactly into three parts. The interesting one asks: do concentrated sellers ship to concentrated buyers? Mildly yes, peaking around 2019, fading through the AI era.

AI compute (3 codes) — joint concentration decomposition (3m MA; dotted = 2023-07, 2025-04)

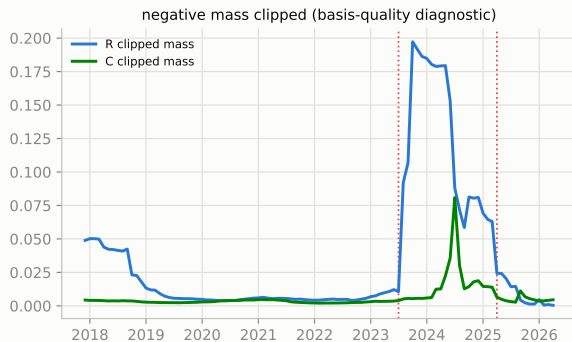
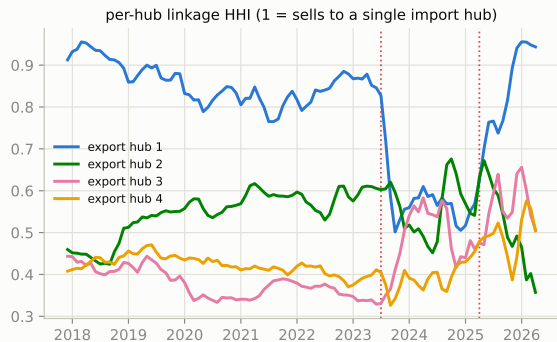


4. Concentration & fragmentation -- AI-compute codes only

One buyer or many?

Left: does each export group sell into one import group or several. Right: a data-quality check, elevated during the 2023-24 transition.

AI compute (3 codes) — linkage structure and basis diagnostics

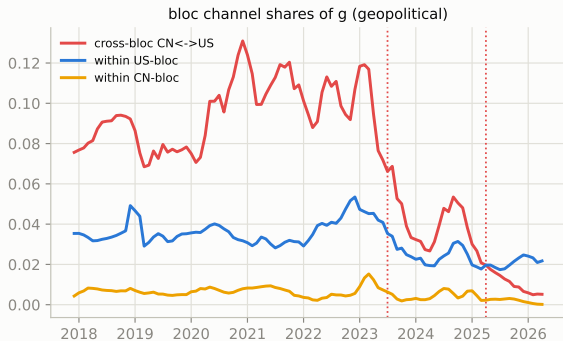
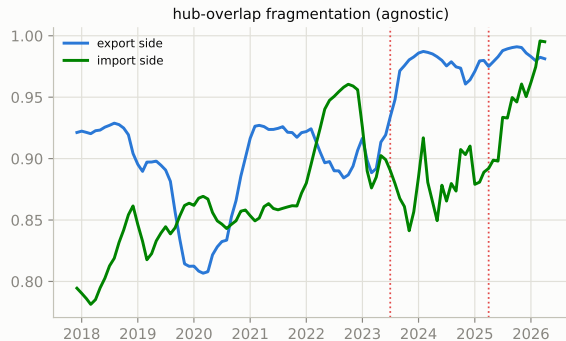


4. Concentration & fragmentation -- AI-compute codes only

Baseboards: the sharpest split

The cross-bloc share collapses about 20x from its 2021 peak -- the code most exposed to export controls.

HS 847180 (baseboards) — two notions of fragmentation (3m MA)



Parts: still shared

The cross-bloc share halves but stays the highest of the three codes: the parts trade remains the layer both blocs still share.

HS 847330 (parts/GPU cards) — two notions of fragmentation (3m MA)

